

Project 3 Final Report

1) Objective

Build an educational end-to-end system to predict treatment outcome (`treatment\_outcome`) from prepared clinical data, then operationalize it with SQL analytics and a local API.

2) Work completed

Completed components:

- EDA, cleaning, and feature engineering.
- Model training/tuning and Scenario 3 comparison.
- XGBoost explainability using SHAP and LIME.
- MySQL connection, full data load, and SQL analytics.
- Flask API for local prediction requests and result logging.

This project follows PDF guidance, adapted to this repository's Scenario 3 assets and MySQL environment.

3) Dataset and Scenario 3 setup

- Prepared patient records: ****1,050,000****.
- Labelled records for supervised modeling: ****985,876****.
- Scenario 3 preprocessing is learned from training split only.
- Split design: ****60/20/20**** (train/validation/test).
- Base predictors used for modeling: ****26****.
- Predictors after feature engineering: ****33****.
- Encoded model inputs: ****121****.

Earlier tuning explored multiple settings and cutoffs. Those tuning outputs are retained as separate experiments and are not merged into the final fixed Scenario 3 comparison.

4) Final model comparison (saved report)

Model selection was based on validation ROC-AUC among Decision Tree, Random Forest, XGBoost, and Neural Network. LSTM is reported as an educational demo.

****Selected model for API use:**** Neural Network (validation ROC-AUC selection rule).

****Important metric highlights:****

- Highest ****test accuracy**** among the five models: ****XGBoost = 72.39%****.
- Neural Network ****test ROC-AUC****: ****0.6496****.

Final test-results table

Model	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC-AUC
Decision Tree	57.95%	0.2997	0.6603	0.4123	0.6459
Random Forest	58.02%	0.2990	0.6541	0.4104	0.6449
XGBoost	**72.39%**	0.3637	0.3154	0.3379	0.6488
Neural Network	59.68%	0.3048	0.6285	0.4105	**0.6496**
LSTM Demo	59.27%	0.3037	0.6373	0.4114	0.6500

Note: LSTM remains educational because each patient contributes one timestep only.

5) Explainability summary (XGBoost)

- SHAP and LIME were implemented for the saved Scenario 3 XGBoost pipeline.
- SHAP provided stable global and local contribution views.
- LIME local fit quality was weak for the sampled case, so LIME outputs are interpreted cautiously.
- These explainability outputs describe model behavior; they do not prove causality.

6) MySQL loading and SQL analytics

- All ****1,050,000**** prepared patients were loaded into ``gsk_medicine_db.patients``.
- Source-to-database verification matched row counts, null counts, and binary outcome distributions.
- Six SQL analytics queries were run from one authoritative SQL source in Python and Workbench.

Selected findings from saved SQL outputs:

1. Outcome distribution: ****72.92% Ineffective****, ****20.97% Effective****, ****6.11% Unknown outcome****.
2. Observed effective rate decreases across age groups in this dataset: about ****37.51% (Under 30)**** vs ****13.67% (Over 65)****.
3. Dosage-category effective rates are close (about ****22.29% to 22.43%****) across low/medium/high groups.

These are observational SQL rates, not causal effects.

7) Flask API integration

- API reads patient predictors directly from MySQL.
- Saved Scenario 3 Neural Network artifacts are reused without retraining.
- Each successful prediction is written to ``prediction_results`` and verified in Workbench joins.

Verified local example:

- ``patient_id``: ****26319****
- Effective score (``probability_effective``): ****46.93%****
- Cutoff: ****50%****
- Predicted class/outcome: ****0 / Ineffective****
- Saved ``prediction_id``: ****1****

8) Clarifications and limitations

- **Model accuracy** is a dataset-level test metric.
- An individual **prediction score** (for one patient) is not model accuracy.
- SQL **observed outcome rates** are descriptive aggregates and not equivalent to model performance.

Limitations:

- The 50% precision target was not met in final comparison.
- LSTM uses one timestep and is educational only.
- The same held-out test split was used in earlier scenarios.
- Deployment is local and educational, not clinical-ready.

9) Conclusion

Project 3 now provides a complete educational workflow: prepared data, compared models, explained XGBoost behavior, verified MySQL analytics, and exposed a local Flask prediction API that stores prediction events. The implementation demonstrates reproducible engineering and reporting while avoiding clinical-readiness claims.